Fjordview Kommune

Fjordview Community Centre

Tender Reference FCC-2026-T01

Document 05

Mechanical, Electrical and Public Health Specification

Revision	Date	Status	Prepared by	Checked by
P01	2026-10-03	Issued for Tender	Fjordview Kommune Project Team	Nordvik Rådgivning AS

This document forms part of the tender documents for the above project. It is issued for tendering purposes only and must not be used for construction.

Note: this is a synthetic example tender pack prepared for a technical interview exercise. The project, parties and figures are fictional.

1. General

1.1 Scope

This Specification covers the mechanical, electrical and public health (MEP) services and the commercial kitchen equipment. The Employer's MEP design is developed to Stage 2 / MMI 300 and is not novated; the Contractor shall appoint its own MEP consultant to complete the design to meet the performance criteria in the Employer's Requirements (Document 03) and this Specification.

Prepared by Nordvik Rådgivning AS. Equipment schedules are indicative of the Employer's intent; the Contractor shall verify all loads, flows and sizes in its completed design.

1.2 Standards

Installations shall comply with NEK 400 (electrical), TEK17, the Norwegian Building Services guides and the standards cited in each section. Reaction-to-fire classification of cables, insulation and ductwork products shall be in accordance with NS-EN 13501-1:2018 and NS-EN 13501-6:2018.

1.3 Submittals

The Contractor shall submit for review, not less than 20 working days before the relevant work commences, the following: product data sheets, samples where requested, method statements, fabrication drawings, and evidence of compliance with the standards cited in this section. Review by the Employer's Representative does not relieve the Contractor of responsibility for compliance with the Contract.

1.4 Quality

Work shall be executed by suitably trained and experienced operatives under competent supervision. The Contractor shall maintain an inspection and test plan for the works in this section and shall give the Employer's Representative not less than two working days' notice of hold points.

1.5 Coordination

The Contractor shall coordinate the MEP services with the architectural and structural design in the federated BIM model and shall resolve all clashes before fabrication. Service zones are shown on the Drawings; clear heights in the Employer's Requirements room data sheets shall be maintained.

2. Heating

2.1 Heat source

Space heating and domestic hot water shall be provided by a ground-source heat pump installation of nominal 180 kW heating output with 12 boreholes of 250 m depth, located under the car park. The municipal district heating connection (Fjordview Fjernvarme) shall be provided as peak and back-up supply with a 150 kW substation in the plant room.

2.2 Distribution

Low temperature hot water at 45/35 degrees C shall be distributed to underfloor heating in all areas at L00 and L01 except the plant room and stores, and to the air handling unit heating coils. Each room shall have individual zone control. Pipework shall be insulated to NS-EN 12828 and labelled.

2.3 Domestic hot water

Hot water shall be generated by the heat pump with a 1,500 l calorifier and electric top-up to 65 degrees C for legionella control. Kitchen hot water shall be at 60 degrees C at the tap.

3. Cooling

Comfort cooling is required only in the Main Hall, Library, Music Room and meeting rooms, provided by the reversible heat pump through the ventilation supply air with chilled beams in meeting rooms. The design cooling load is 95 kW. Server and comms rooms shall have dedicated split units with N+1 redundancy.

4. Public health

Cold water shall be supplied from the municipal main via a 50 mm connection with a break tank and booster set in the plant room. Low-flow fittings shall be used throughout to meet the water consumption target in the Employer's Requirements. Foul drainage shall discharge by gravity to the public sewer in Havnegata. Kitchen drainage shall pass through a grease separator (NS-EN 1825, nominal size 4) located in the service yard. Rainwater shall be collected to the attenuation tank and discharged at the limited rate stated in the Employer's Requirements.

5. Electrical services

5.1 Incoming supply

A new 400 V three-phase supply shall be taken from the BKK Nett network in Havnegata to a main switchboard in the roof-level plant room. The Contractor shall make the supply application and pay connection charges under provisional sum PS5. The main switchboard shall have 25% spare capacity in both physical space and rating.

No on-site generation is included in the scope of this Specification. The main switchboard shall be provided with a spare 100 A outgoing way for future connection of additional loads.

5.2 Maximum demand

The assessed maximum demand of the building is 400 kW (approximately 580 A per phase at 0.95 power factor) based on the loads in the following sections with diversity. The Contractor shall confirm the maximum demand in its completed design.

5.3 Distribution boards

Board	Location	Serves	Rating
MSB	Plant room RF	All sub-boards	630 A
DB-L00-1	Electrical riser L00	Foyer, cafe, MPRs, changing	160 A
DB-L00-2	Main Hall store	Main Hall power and stage	250 A

Board	Location	Serves	Rating
DB-K1	Kitchen L00-021	Commercial kitchen equipment	250 A
DB-L01-1	Electrical riser L01	Library, Music Room, offices	160 A
DB-L01-2	Youth Club store	Youth Club, meeting rooms	100 A
DB-M	Plant room RF	Mechanical plant	250 A
DB-EXT	Plant room RF	External lighting, EV charging	160 A

5.4 Kitchen distribution board DB-K1 - connected loads

The following connected loads shall be allowed for at DB-K1. Final loads shall be confirmed against the selected equipment in Section 7.

Circuit	Equipment tag	Description	Connected load	Supply
1-3	OV-01	Combi oven, 10-grid	18 kW	3-phase
4-6	OV-02	Convection oven	6.5 kW	3-phase
7-9	HB-01	Induction hob, 4-zone	14 kW	3-phase
10-12	FZ-01	Walk-in freezer	40 kW	3-phase
13-15	CH-01	Walk-in chiller	2.5 kW	3-phase
16	MW-01	Microwave oven	1.5 kW	1-phase
17-19	DW-01	Pass-through dishwasher	9.5 kW	3-phase
20	FR-01	Fryer, single tank	9 kW	3-phase
21	BM-01	Bain-marie	2 kW	1-phase
22	RF-01	Upright refrigerator	0.4 kW	1-phase
23	RF-02	Upright refrigerator	0.4 kW	1-phase
24	HD-01	Hand dryer, kitchen staff change	1.4 kW	1-phase
25-30	-	Small power sockets	6 kW	1-phase

5.5 Lighting

Lighting shall be LED throughout with DALI control, presence detection and daylight dimming in all spaces with windows, to achieve the illuminance levels in the Employer's Requirements Section 6.3. Lighting power density shall not exceed 6 W/m2. Emergency lighting shall be self-contained luminaires with 3-hour duration and automatic testing.

5.6 Small power and containment

Small power shall be provided to the room data sheets with a minimum of two twin socket outlets per workstation and one twin outlet per 10 m2 in public rooms. Containment shall be galvanised steel tray and basket in risers and ceiling voids, with 30% spare capacity.

5.7 EV charging

Eight 22 kW Mode 3 charging points shall be provided in the car park with load management, supplied from DB-EXT, with ducting to a further 16 bays.

6. Ventilation

6.1 Air handling

Four air handling units shall be provided in the roof plant room, each with rotary heat recovery of not less than 85% temperature efficiency, EC fans and F7 filtration: AHU-1 Main Hall (8,500 m3/h), AHU-2 L00 general (6,000 m3/h), AHU-3 Library and L01 general (7,000 m3/h), AHU-4 kitchen supply (4,000 m3/h). Kitchen extract shall be a separate system with a UV-C grease treatment canopy and discharge above roof level.

6.2 Design ventilation rates

Occupancy ventilation rates for the completed design shall not be less than the following:

Space	Design occupancy	Rate per person	Control
Main Hall	600	12 l/s	CO2 demand control
Library	120	10 l/s	CO2 demand control
Multi-purpose rooms	40 each	10 l/s	CO2 demand control
Music Room and practice rooms	30 / 4	10 l/s	Presence
Offices and meeting rooms	24 / 12	8 l/s	Presence
Youth Club	60	10 l/s	CO2 demand control
Cafe	60	10 l/s	Presence
Kitchen	8	Per canopy and equipment heat gain	Constant

Supply air temperature shall be variable between 16 and 21 degrees C. Specific fan power shall not exceed 1.5 kW/(m3/s) as required by the Employer's Requirements.

6.3 Smoke control

Smoke ventilation to the Main Hall and the Library shall be provided by automatic opening vents sized by the Contractor's fire engineer.

7. Commercial kitchen equipment

7.1 General

The Contractor shall supply, install, connect and commission the equipment scheduled below, procured through a specialist Supplier with not less than 5 years' experience of commercial kitchens. Equipment shall be stainless steel grade 304, CE marked, and shall carry the warranty required by the Particular Conditions of Contract Clause 11.3. The equipment schedule and the DB-K1 schedule in Section 5.4 shall be read together; the Contractor shall reconcile any difference before ordering.

7.2 Equipment schedule

Tag	Description	Qty	Connected load	Electrical supply	Services
OV-01	Combi oven, 10-grid GN1/1, boiler type	1	18 kW	3N 400 V, 32 A	Cold water, drain
OV-02	Convection oven, 6-grid	1	6.5 kW	3N 400 V, 16 A	-
HB-01	Induction hob, 4-zone, on cupboard base	1	14 kW	3N 400 V, 32 A	-
FR-01	Fryer, single tank 15 l	1	9 kW	3N 400 V, 16 A	-

Tag	Description	Qty	Connected load	Electrical supply	Services
BM-01	Bain-marie, 3 x GN1/1	1	2 kW	230 V, 13 A	-
MW-01	Microwave oven, commercial 1,000 W output	1	1.5 kW	230 V, 13 A	-
FZ-01	Walk-in freezer, 2.4 x 3.0 x 2.2 m, -20 degrees C, remote condensing unit on roof	1	4.0 kW	3N 400 V, 16 A	Condensate drain
CH-01	Walk-in chiller, 2.4 x 3.0 x 2.2 m, +3 degrees C, remote condensing unit on roof	1	2.5 kW	3N 400 V, 16 A	Condensate drain
RF-01	Upright refrigerator 650 l	1	0.4 kW	230 V, 13 A	-
RF-02	Upright refrigerator 650 l	1	0.4 kW	230 V, 13 A	-
DW-01	Pass-through dishwasher with heat recovery	1	9.5 kW	3N 400 V, 16 A	Hot water, drain
ST-01	Stainless steel preparation table 1,800 mm	4	-	-	-
ST-02	Stainless steel wall shelf 1,200 mm	8	-	-	-
SK-01	Double bowl sink with drainer	2	-	-	Hot and cold water, drain
SK-02	Hand wash basin, knee operated	2	-	-	Hot and cold water, drain
CN-01	Extract canopy 4.0 x 1.4 m with UV-C and lighting	1	0.6 kW	230 V	Extract duct
SV-01	Servery counter heated display, 1,500 mm	1	3 kW	230 V, 16 A	-
SV-02	Servery counter refrigerated display, 1,500 mm	1	0.8 kW	230 V, 13 A	Condensate drain

7.3 Installation

Walk-in cold rooms FZ-01 and CH-01 shall be modular insulated panel construction with floors, supplied complete with doors (door references D-G16 and D-G17 in the Architectural Specification), lighting, temperature monitoring with remote alarm and remote condensing units on the roof within the plant screen. Refrigerant shall be R290 or CO2.

8. Fire detection and alarm

An addressable analogue fire detection and alarm system to NS 3960 category 2 shall be provided with voice alarm in the Main Hall and Library, interfaced with the hold-open devices, smoke vents, lifts and ventilation shutdown. The system shall be connected to the fire brigade via an approved alarm transfer service.

9. Lifts

One (1) machine-room-less passenger lift shall be provided, rated 1,000 kg / 13 persons, 1.0 m/s, serving L00, L01 and the roof plant level, with a car size of 1,100 x 2,100 mm to accommodate a stretcher and conforming to NS-EN 81-70 for accessibility. The lift shall be designated as the firefighting lift and shall comply with NS-EN 81-72, with a secondary power supply.

A goods lift is not required.

10. Building management system

A BMS shall monitor and control all mechanical plant, lighting and energy metering with an open protocol (BACnet/IP) and web access for the Employer's facilities team. Sub-metering shall be provided by end use to allow the energy label and the net delivered energy target to be verified in operation.

11. Testing, commissioning and handover

All systems shall be tested and commissioned in accordance with the relevant standards and the Particular Conditions of Contract Clause 10. Commissioning results shall be witnessed by the Employer's Representative. At completion the Contractor shall provide operation and maintenance manuals, as-built drawings in native and PDF format, commissioning records, warranties and training for the Employer's facilities staff.